When Does Private Equity Pay Off? US Private Equity vs Public Markets

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Technical terms are defined in the Glossary at the end of this note.

Executive summary

US Private Equity has beaten public equities over the long run, but the premium is modest once measured properly and it depends far more on when and with whom an investor commits, than on the asset class itself.

- **Long run:** over 25 years, the Cambridge Associates US PE Index returned 12.76% a year (to Mar 2026), against 8.74% for the S&P 500 and 6.18% for Baa corporate bonds (to Dec 2025).
- **Measurement matters:** measured like-for-like (mPME vs the Russell 3000), the 25-year premium is +273 bps a year; a naive comparison with the S&P 500 suggests +402 bps, partly because of the method and partly because of the benchmark. Over 1 to 5 years, PE underperformed US public equities (Russell 3000, mPME) by 228 to 1,196 bps a year.
- **Vintage matters:** funds raised after the dot-com crash (2001–2003) beat the S&P 500 mPME by c. 1,240 bps a year; funds raised at the pre-GFC peak (2005–2008) slightly lagged it (c. −40 bps).
- **The manager matters most:** within the same vintage, the gap between top- and bottom-quartile funds averaged 14.1 points a year across 1994–2013 vintages, about 2.5 times the average PE value-add over the S&P 500 (5.5 points). For an investor, choosing the right manager matters more than the decision to invest in PE at all.
- **What this suggests:** over 25 years, US PE as a whole outperformed US public equities by about 2.7 points a year, just below the c. 3-point margin that many pension funds use as their PE benchmark. The results are consistent with common investor practice: spreading commitments evenly across vintage years rather than timing the market, and treating manager selection as the key decision.

1. Question and approach

The question for an investor is not whether PE returns are high, but whether they beat what the same money would have earned in liquid markets. PE asks for a 10-year lock-up, high fees and limited transparency, so it must earn a premium over listed assets.

This note compares US private equity with US equities and bonds along three dimensions:

1. **Horizon:** annualised returns over 1 to 25 years.
2. **Vintage year:** performance by the year a fund started investing (1994–2013).
3. **Manager:** the spread between top- and bottom-quartile funds.

2. Data and methodology

The analysis combines three public datasets, rebuilt in an Excel model and cross-checked in Python with identical results.

Dataset	Content	As of
Cambridge Associates, US PE Benchmark Book Q1 2026	Pooled horizon IRR of 1,801 US private equity funds formed 1983–2026; mPME vs MSCI World/ACWI (constructed), Russell 3000 and Russell 2000	31 Mar 2026
Cambridge Associates, US PE (legacy definition) Benchmark Book Q3 2019	Since-inception IRR, quartiles and mPME by vintage year, 1,366 funds (1994–2017)	30 Sep 2019
A. Damodaran, NYU Stern, Historical Returns	Annual total returns of the S&P 500, US small caps, T-bills, 10Y T-bonds and Baa bonds (1928–2025)	31 Dec 2025

Public assets are measured by their geometric annualised return (CAGR), which compounds the annual returns over the last *n* years:

CAGR = $[(1 + r_1) \times (1 + r_2) \times \ldots \times (1 + r_n)]^{(1/n)} - 1$, where r_t is the total return in year t.

PE is measured by its internal rate of return (IRR), net of fees, expenses and carried interest. The IRR is money-weighted: it depends on when capital is called and returned. Comparing it directly with a time-weighted CAGR is therefore *naive*.

The modified Public Market Equivalent (mPME) corrects this. Cambridge Associates invests each fund's capital calls in a public index at the same dates, and sells the same proportion of holdings at each distribution. The PE *value-add* is the fund's IRR minus the IRR of this shadow portfolio: a like-for-like comparison.

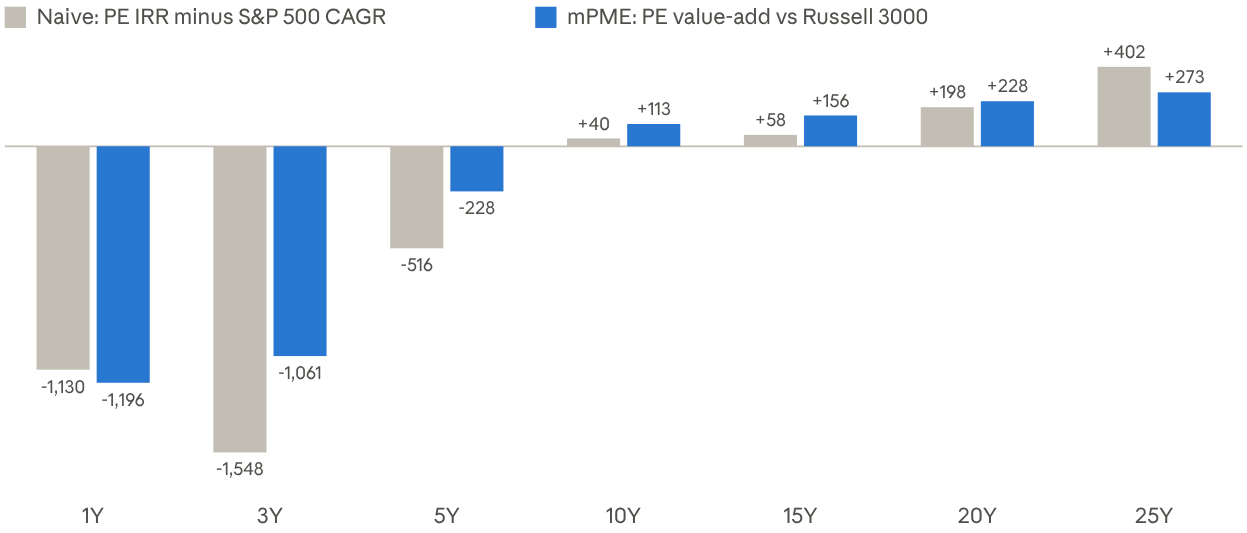
3. Findings

3.1 PE wins over the long run, but the measure changes the size of the win

Over 25 years, PE returned 12.76% a year, against 8.74% for the S&P 500, 9.65% for US small caps and 6.18% for Baa bonds. Over 1 to 5 years the ranking flips: the S&P 500 compounded at 14 to 23% a year, PE at 6 to 9%.

Like-for-like, PE beats US equities only over 10 years or more

Over 25Y: +273 bps a year with the mPME, vs +402 bps with the naive comparison



Cambridge Associates US PE Index Q1 2026 (PE IRR and mPME, to 31 Mar 2026); A. Damodaran, NYU Stern (S&P 500 CAGR, to 31 Dec 2025)

In this chart the naive gap and the mPME use different indices (S&P 500 vs Russell 3000), so part of the difference comes from the benchmark. The 2019 report isolates the effect of the method alone, against the same index: the naive comparison misstates the premium by up to 1.5 points a year, in either direction.

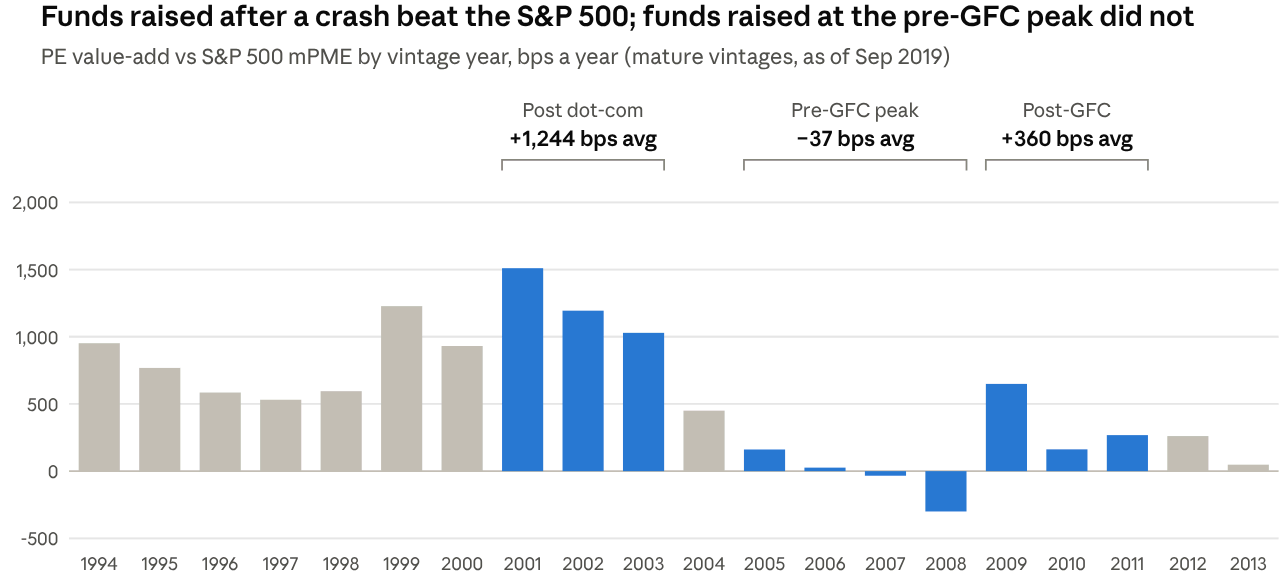
Horizon (to 30 Sep 2019)	Naive gap vs S&P 500	mPME value-add vs S&P 500	Naive minus mPME
15 years	+443 bps	+400 bps	+43 bps

Horizon (to 30 Sep 2019)	Naive gap vs S&P 500	mPME value-add vs S&P 500	Naive minus mPME
20 years	+538 bps	+390 bps	+148 bps
25 years	+352 bps	+431 bps	−79 bps

Source: Cambridge Associates, US PE (legacy definition) Benchmark Book Q3 2019, pp. 7–8.

3.2 The vintage year matters: post-crash funds outperformed

A fund invests most of its capital in the 3 to 5 years after launch, so its returns depend on the prices it pays. Funds raised after the dot-com crash bought cheaply and beat the S&P 500 mPME by c. 1,240 bps a year; funds raised at the 2005–2008 peak roughly matched it.

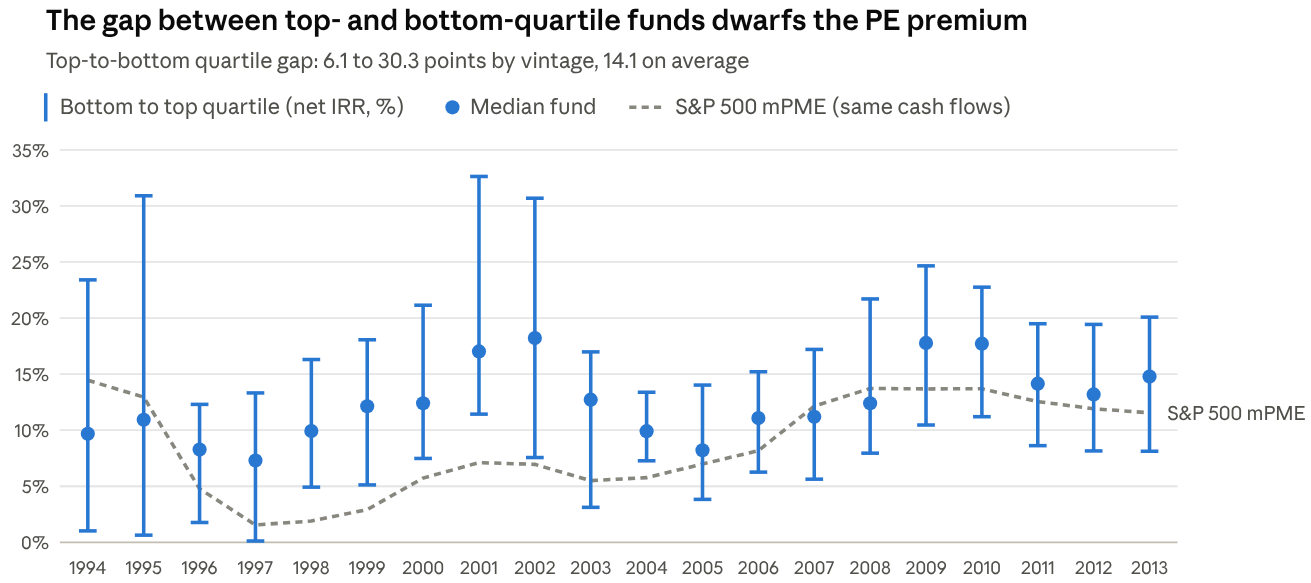


Cambridge Associates, US PE (legacy definition) Benchmark Book Q3 2019: since-inception pooled IRR vs S&P 500 mPME, vintages 1994–2013, as of 30 Sep 2019

The value-add is largest when public markets are weak over the fund's life. For 1997–2003 vintages, the S&P 500 equivalent earned only 2 to 7% a year; for 2007–2010 vintages, which lived through the post-2009 bull market, it earned 12 to 14%.

3.3 The manager matters more than the asset class

Across mature vintages, the gap between top- and bottom-quartile funds averaged 14.1 points a year. The average pooled value-add over the S&P 500 mPME was 5.5 points.



Cambridge Associates, US PE (legacy definition) Benchmark Book Q3 2019: since-inception net IRR by vintage, 1994–2013, as of 30 Sep 2019

In 14 of 20 vintages, a bottom-quartile fund returned less than the pooled S&P 500 equivalent. Choosing the manager is therefore a bigger decision than allocating to PE at all.

4. What this means for investors

The results do not suggest that PE is attractive in all conditions, but they point to three factors that drive whether it pays off for an investor :

1. **Timing of commitments.** The best vintages followed market crashes, when many investors cut back on new commitments. Since crashes cannot be predicted, committing a steady amount every year (vintage diversification) captures these vintages without having to time the market.
2. **Manager selection.** Across 1994–2013 vintages, the gap between top- and bottom-quartile funds averaged 14.1 points a year, about 2.5 times the average PE value-add over the S&P 500 (5.5 points). Selection is hard to do in advance, however: for buyout funds raised since 2000, past top-quartile performance has been a weak predictor of future results (Harris et al., 2020).
3. **Measurement.** Even against the same index, a naive IRR-versus-index comparison can misstate the PE premium by up to 1.5 points a year, in either direction depending on the period. This supports the use of PME-based measures when assessing a manager's track record.

5. Limitations

The conclusions rest on public, aggregated data and should be read with seven caveats:

- **Valuation smoothing:** PE NAVs are quarterly appraisals, not market prices. Recent returns (1 to 5 years) depend heavily on these estimates, and reported PE volatility is understated.
- **No risk adjustment:** buyouts are leveraged. Part of the premium may reward extra risk rather than manager skill.
- **Data access:** the latest free vintage data dates from September 2019, so vintages after 2013 are excluded as immature. Some vintages hold only 31 to 35 funds (2001–2003 and 2009–2010).
- **Definitions differ:** the 2019 data use CA's legacy definition (including mezzanine and energy funds) and define vintage by legal inception; CA's current index (2026) covers buyout and growth funds only.
- **Timing mismatch:** PE horizon returns run to March 2026, public returns to December 2025.
- **Benchmark mismatch:** the 2026 report gives the mPME vs the Russell 3000 but not vs the S&P 500, while annual public returns are only available for the S&P 500. Naive and mPME figures therefore differ by both method and index.
- **Reporting bias:** CA's database relies on funds reporting their results. Funds can be added or removed over time, so past returns may change between editions.

Sources

- Cambridge Associates, US Private Equity Benchmark Book, Q1 2026, as of 31 Mar 2026.
- Cambridge Associates, US Private Equity (Legacy Definition) Benchmark Book, Q3 2019, as of 30 Sep 2019.
- A. Damodaran, Historical Returns on Stocks, Bonds and Bills, 1928–2025, NYU Stern, updated 5 Jan 2026.
- Harris, Jenkinson, Kaplan & Stucke, *Has Persistence Persisted in Private Equity? Evidence from Buyout and Venture Capital Funds*, NBER Working Paper 28109, 2020.
- CalPERS, Private Equity benchmark review, Investment Committee, April 2017.
- Institutional Investor, *Public pensions frequently change their private equity benchmarks. Why it matters*.

Glossary

Players

- **Private equity (PE):** investment in companies that are not listed on a stock exchange, usually by buying a majority stake, improving the business and selling it a few years later.
- **Buyout (LBO):** the purchase of a company financed partly with debt, to increase the return on the money invested.
- **Fund:** a pool of money raised by a PE firm for a limited life (about 10 years), used to buy and later sell several companies.
- **GP (General Partner):** the PE firm that manages the fund and chooses the investments (e.g. KKR, Ardian). Also called the *manager*.
- **LP (Limited Partner):** an investor that puts money into a PE fund (e.g. a pension fund, an insurer).
- **Vintage year:** the year a fund starts investing. Funds of the same vintage invest in the same market conditions.
- **Capital call / distribution:** a capital call is money a fund requests from its LPs to make an investment; a distribution is cash a fund returns to its LPs, usually after selling a company.
- **Carried interest:** the share of a fund's profits (usually about 20%) kept by the GP when the fund performs well.

Returns

- **IRR (Internal Rate of Return):** the annual return of a PE fund, taking into account *when* money is invested and returned. Also called a *money-weighted* return.
- **CAGR (Compound Annual Growth Rate):** the average annual return of an investment held throughout a period, with gains reinvested. Also called a *time-weighted* return.
- **Pooled return:** the combined return of all funds in a group, as if they were one large fund. Bigger funds weigh more.
- **Basis point (bp):** one hundredth of a percentage point. 100 bps = 1 point; 273 bps = 2.73 points.
- **Quartile:** one of four equal groups when funds are ranked from best to worst. *Top quartile* = the best 25%; *bottom quartile* = the worst 25%.

- **NAV (Net Asset Value):** the estimated value of the companies a fund still owns. It is an appraisal, not a market price.

Comparing PE with public markets

- **Public markets:** stocks and bonds traded on an exchange, which can be bought or sold at any time.
- **S&P 500:** an index of the 500 largest US listed companies.
- **Russell 3000:** an index of the 3,000 largest US listed companies, covering almost the whole US stock market.
- **Russell 2000:** an index of the 2,000 smallest companies in the Russell 3000.
- **US small caps (bottom decile):** the smallest 10% of US listed companies by market value, as used in Damodaran's data.
- **Baa corporate bonds:** bonds issued by US companies with a medium credit rating (Moody's Baa), just above high-yield status.
- **T-bills / 10Y T-bonds:** US government debt maturing in 3 months / 10 years.
- **Naive comparison:** subtracting a stock index's CAGR from a PE fund's IRR. Simple but misleading, because the two returns are not measured the same way.
- **PME (Public Market Equivalent):** a method that invests a PE fund's cash flows in a stock index, on the same dates, to compare like with like.
- **mPME (modified PME):** Cambridge Associates' version of the PME, which sells the same *proportion* of the index at each fund distribution, so the shadow portfolio never turns negative.
- **Value-add:** the PE fund's IRR minus the return of its mPME. A positive value-add means PE beat the stock market.

Context

- **Illiquidity premium:** the extra return investors expect for locking up their money for years. Many pension funds expect PE to beat stock markets by about 3 points a year.
- **Dot-com crash (2000–2002):** the collapse of internet and technology stocks after a speculative bubble.
- **GFC (Global Financial Crisis, 2007–2009):** the financial crisis triggered by US subprime mortgages, marked by the failure of Lehman Brothers in 2008.